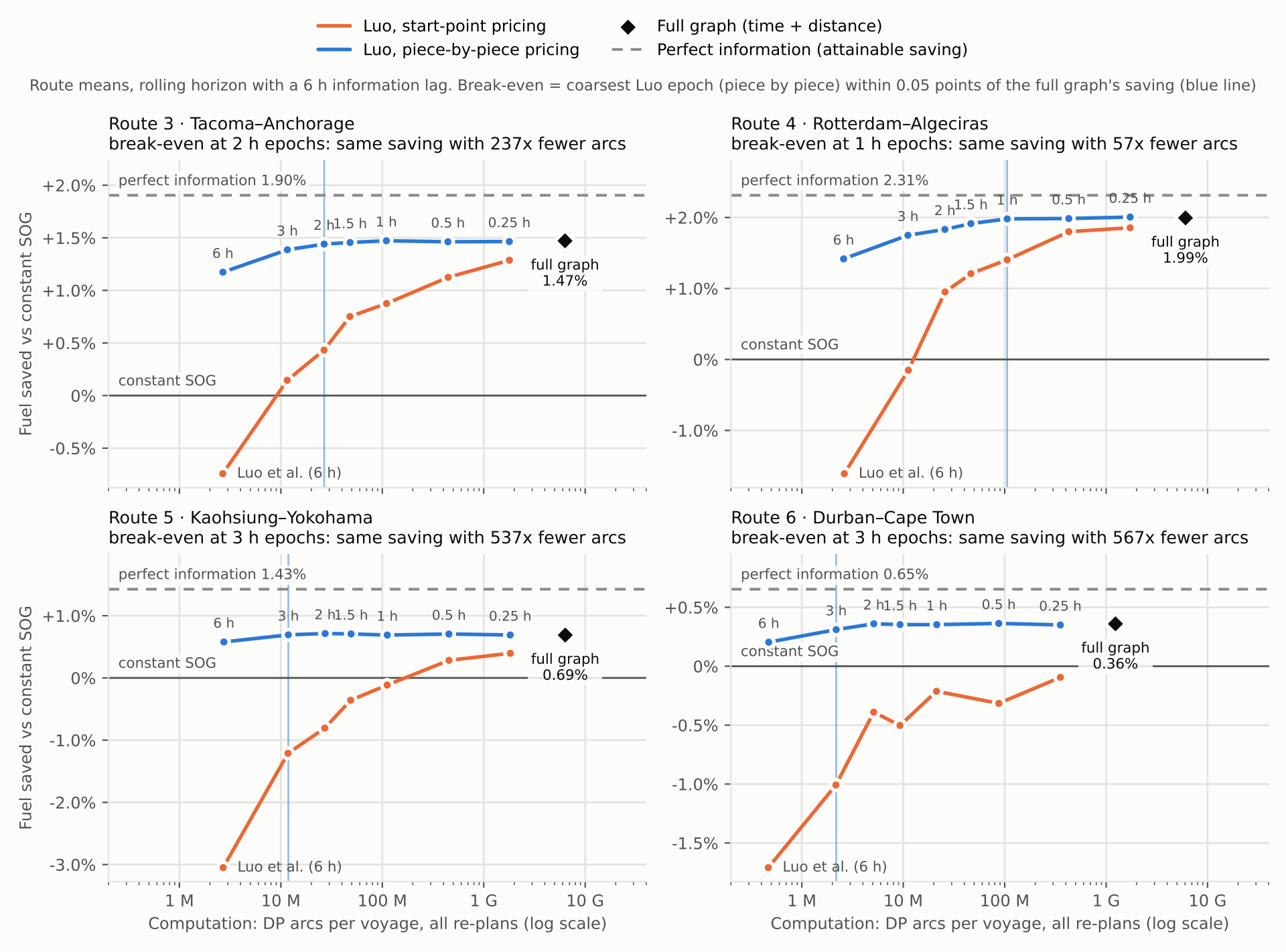
Luo, start-point pricing
Luo, piece-by-piece pricing
Full graph (time + distance)
Perfect information (attainable saving)
Route means, rolling horizon with a 6 h information lag. Break-even = coarsest Luo epoch (piece by piece) within 0.05 points of the full graph's saving (blue line)
Route 3 · Tacoma–Anchorage
break-even at 2 h epochs: same saving with 237x fewer arcs
perfect information 1.90%
full graph 1.47%
Route 4 · Rotterdam–Algeciras
break-even at 1 h epochs: same saving with 57x fewer arcs
perfect information 2.31%
full graph 1.99%
Route 5 · Kaohsiung–Yokohama
break-even at 3 h epochs: same saving with 537x fewer arcs
perfect information 1.43%
full graph 0.69%
Route 6 · Durban–Cape Town
break-even at 3 h epochs: same saving with 567x fewer arcs
perfect information 0.65%
full graph 0.36%
6 h
3 h
2 h
1.5 h
1 h
0.5 h
0.25 h
constant SOG
Luo et al. (6 h)
Fuel saved vs constant SOG
1 M
10 M
100 M
1 G
10 G
Computation: DP arcs per voyage, all re-plans (log scale)